

End Term Examination of Spring Semester 2024

(For 1st Year 2nd Semester of 3-Year BCA)

Subject: Data Structure & Algorithms

Paper Code: BCAUGMCC1201

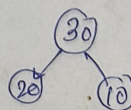
Full Marks: 80

Time: 3 hours

Group A
(Answer all)

1 X 10 = 10

- 1.
- a) Best case of insertion in an array is
(i) to insert at beginning of the array.
(ii) to insert at end of the array.
(iii) to insert at middle of the array
(iv) both (i) and (ii).
- b) Which of the following is not the required condition for binary search algorithm?
(i) The list must be sorted **Y**
(ii) There should be the direct access to the middle element in any sub-list
(iii) There must be mechanism to delete and/or insert elements in list
(iv) none of above
- c) The postfix expression for the infix expression $A + B * (C + D) / F + D * E$ is:
(i) $AB + CD * F D + E *$
(ii) $A * B + CD F * DE + +$
(iii) $ABCD * F DE * + +$
(iv) $A + * BCD F * DE + +$
 $A + B * (CD) / F + D * E$
 $A + BCD + * / F + DE *$
 $A + BCD + * F / + DE * +$
- d) Which of the following is not the internal sort?
(i) Insertion Sort
(ii) Selection Sort
(iii) Merge Sort
(iv) Quick Sort
- e) Binary search performs worse than sequential search when
(i) searching is unsuccessful.
(ii) searching element is present at middle of the list.
(iii) searching element is present at end of the list.
(iv) searching element is present at beginning of the list.
- f) Which data structures are used in case of DFS and BFS graph traversal algorithm?
(v) DFS Stack and BFS Stack
(vi) DFS Queue and BFS Stack
(vii) DFS Queue and BFS Queue
(viii) None
- g) The following sequence of operations is performed on a stack:
PUSH (10), PUSH (20), POP, PUSH (10), PUSH (20), POP, POP, POP, PUSH (20), POP.
The sequence of values popped out is:
(i) 20, 20, 10, 10, 20
(ii) 20, 10, 20, 10, 20
(iii) 10, 20, 20, 10, 20
(ix) 20, 20, 10, 20, 10
10, 20
- h) The Preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42. Which one of the following is the Post-order traversal sequence of the same tree?
(i) 10, 20, 15, 23, 25, 35, 42, 39, 30
(ii) 15, 10, 25, 23, 20, 42, 35, 39, 30
(iii) 15, 20, 10, 23, 25, 42, 35, 39, 30
N L R



(iv) 15, 10, 23, 25, 20, 35, 42, 39, 30

- i) The way a card game player arranges his cards as he picks them up one by one is an example of
- Bubble sort
 - Selection Sort
 - Insertion Sort
 - Radix sort

- j) A binary search tree that is generated by inserting in order the following integers:

50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24

The number of nodes in the left sub-tree and right sub-tree of the root is respectively is

- (i) (4, 7) (ii) (7, 4) (iii) (8, 3) (iv) (3, 8)

Group B

(Answer any five)

2.

- Write algorithm to insert a new element in an existing array. Explain it with suitable example.
- Insertion and deletion in linked list are easier than that in array; explain with suitable example.
- What is polish notation? What are its advantages and assumptions.
- Show the steps of bubble sort to sort the list: 12, 78, 65, 47, 6, 87, 3.
- Define tree. What do you mean by tree traversal? Why tree traversal algorithm is limited to three only?
- Is quick sort is better than merge sort? Justify your answer. What is pivot? How to choose pivot?
- Differentiate liner and non - linear data structures. Give one example for each.

$$5 \times 5 = 25 \\ 3 + 2$$

$$1 + 1 + 3 \\ 1 + 2 + 2 \\ 3 + 2$$

Group B

(Answer any three)

$$15 \times 3 = 45$$

- Write an algorithm to insert a new node in an existing single linked list for which minimum steps are required.
- Differentiate Array and single linked list.
- List few applications of stack. Evaluate the following expression using Stack: 15, 6, 3, +, *, 12, 4, /, -

$$5+5+5$$

- Define recursion. What is its problem over iteration? When to use recursion? Write a recursive function for factorial.
- What is hashing? Briefly discuss common Hash function. What is the problem of hashing? How this problem can be removed?
- Write a recursive algorithm to solve Tower of Hanoi problem.

$$6+6+3$$

- What are the advantages of balance tree over unbalanced tree? Explain with example.
- Write the steps to insert a new element in B-tree.
- Define BST. Write down the steps to delete any node from BST.
- Write the algorithm to merge two sorted arrays into a single array.

$$2+4+4+5$$

- What do mean by "comparison-based sorting techniques" and "non-comparison-based sorting techniques"? Give example
- Write an algorithm to delete any particular element from array.
- What is PUSH? Write algorithm for POP.
- Construct the BST tree with the following elements: 25, 57, 38, 47, 12, 92, 86, 33, 1, 4, 3.

$$2+3+4+6$$

- Following two traversal sequences are given. Construct the corresponding Binary Tree:

In-order: E A C K E H D B G

Preorder: F A E K C D H G B

- What are different representations of Graphs in memory?

- Find out the Adjacency Matrix and Adjacency List representation of the following undirected Graph:

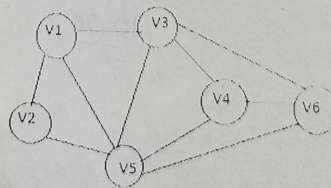


Fig. 1

$$6+3+6$$

